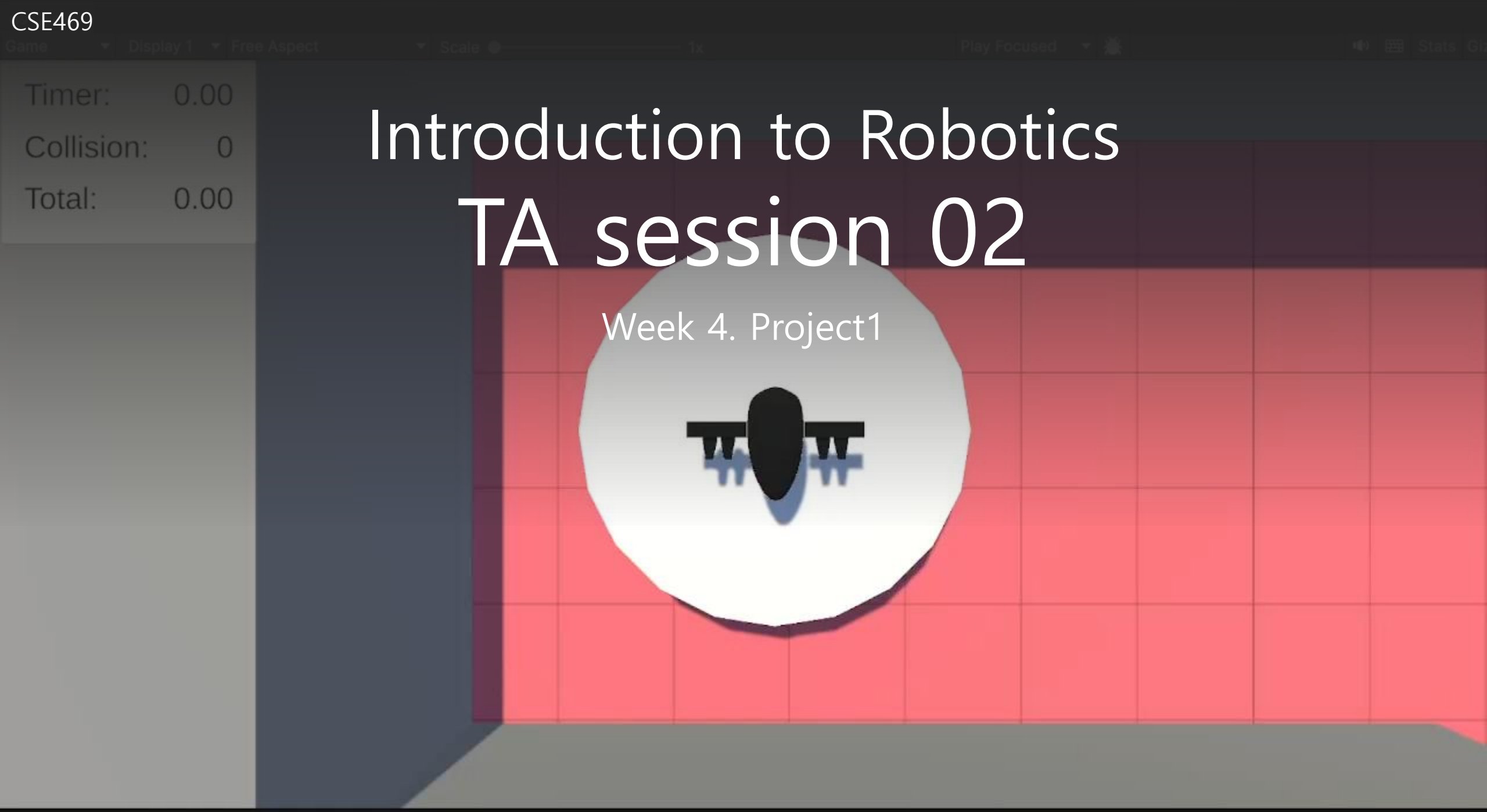
CSE469

Introduction to Robotics

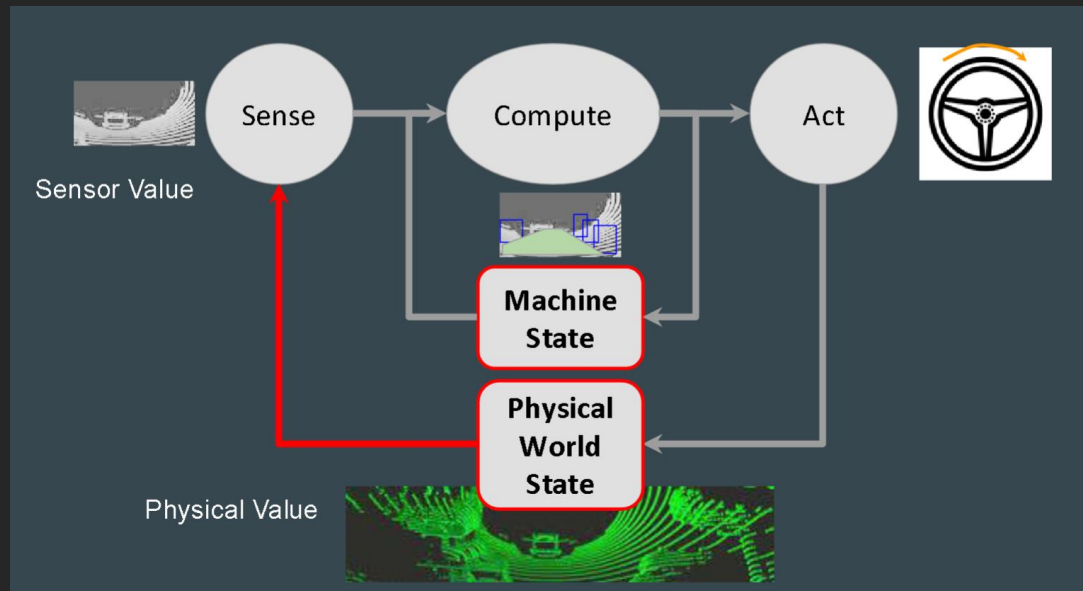
TA session 02

Week 4. Project1



What we've learned so far…

- Robot Architecture (L02)
  - Sense, Compute, Act
- Basic Components (L03)
  - Resistor, Capacitor, Inductor
  - Diode, Transistor etc…
  - Circuit construction tools
- Arduino (TA01)
- Noise Management (L05)
  - Calibration
  - Filtering
  - Fusing



Time for a Project!

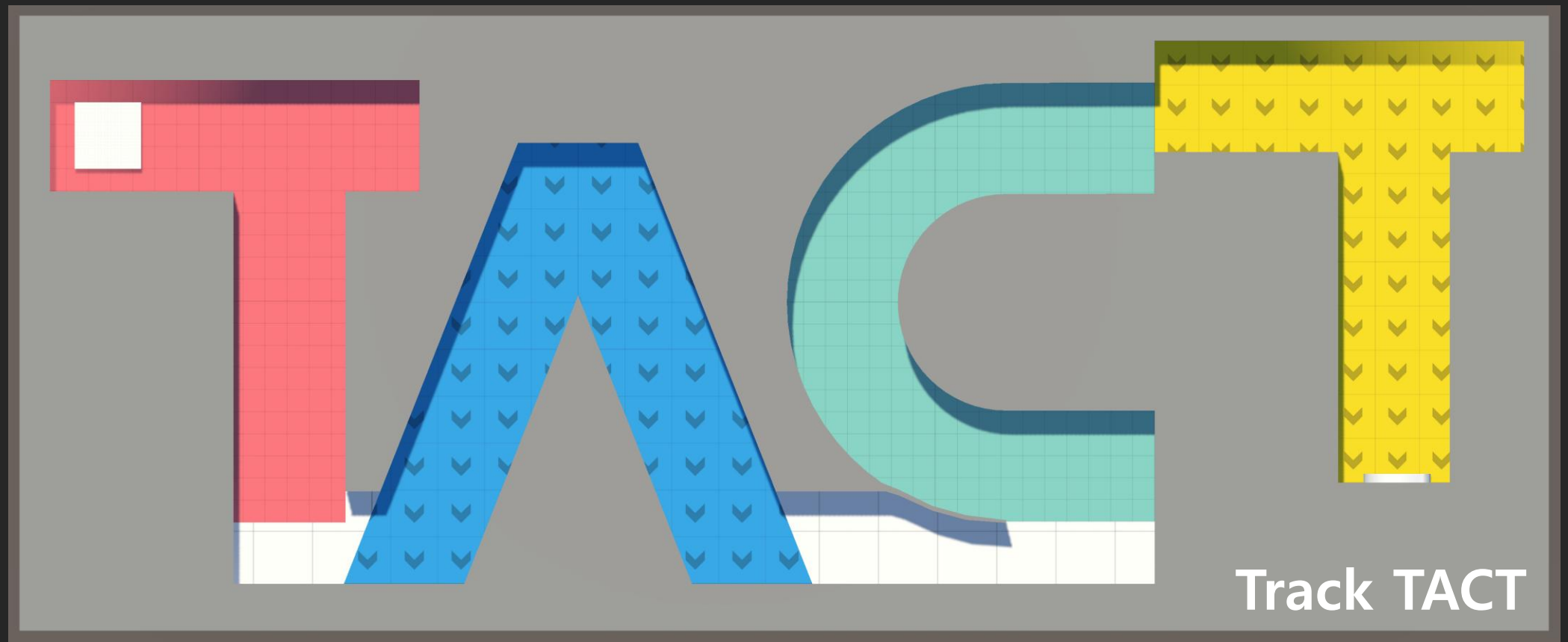
- Your final mission is to mine rare resources on an alien planet.
- To help you on that journey, it will be divided into three projects.

Project Plan

- Project 1: Teleoperation
  - Building a spaceship controlled by human, without sensors.
- Project 2: Autonomy
  - Building a fully autonomous, sensor-based drones.
- Project 3: Shared Autonomy
  - Building a robotic system for
    - a human-operated mothership
    - and multiple autonomous resource-gathering drones.

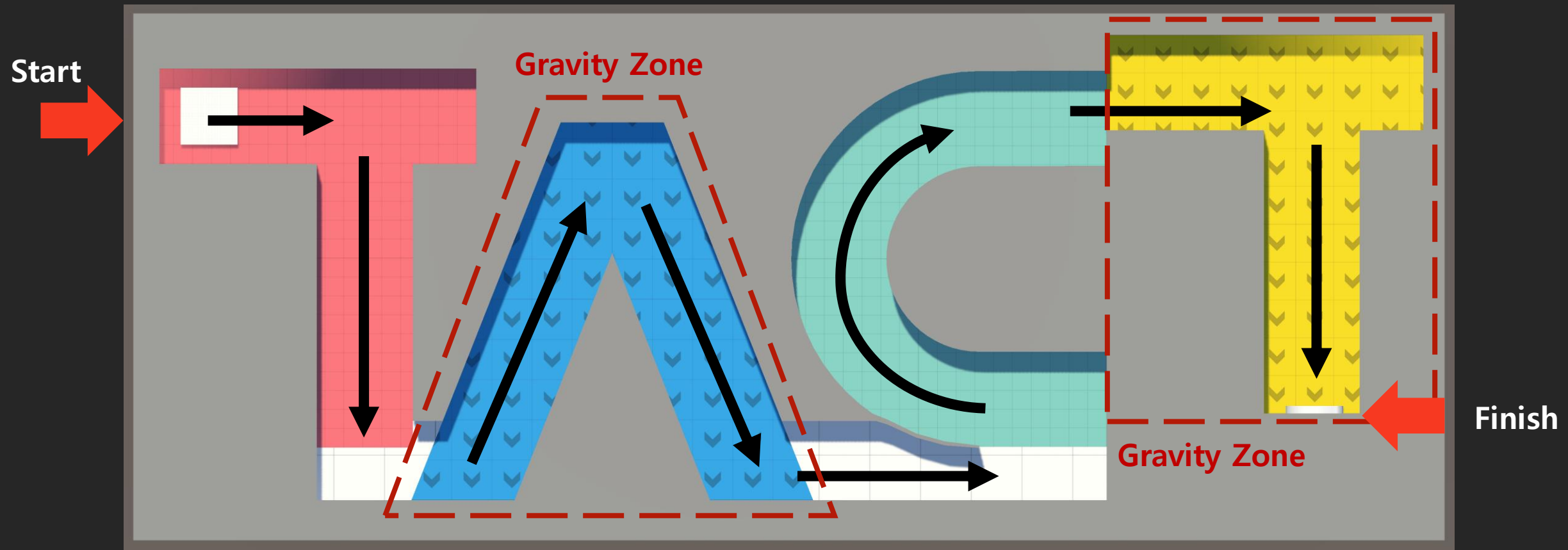
Project 1: Teleoperation

- Design a **spacecraft**, **controller**, and **control system** to reach its destination as quickly and safely as possible on a given track.



Project 1: Teleoperation

- Design a **spacecraft**, **controller**, and **control system** to reach its destination as quickly and safely as possible on a given track.



Files to be submitted

- Performance report
  - Performance record
  - Participation of each team members.
    - Ex) Design of spacecraft, controller, control system and design considerations
- Performance video
  - Proof of your performance report.
  - full view of Unity screen, controller and hands, recorded from starting line to finish line.
- Resource file Unity package
  - Must contain final design of control system and spacecraft.
- If manipulation is suspected in the submitted file, additional session may be required to demonstrate the control.

Evaluation criteria

- Performance Report (10 points, Team score)
  - Final Time = Time + Number of Collisions * 5 seconds
  - If a collision occurs, no further collisions will be detected for 5 seconds.
    - 200 seconds or more: 2 points
    - 200-170: 4 points
    - 170-140: 6 points
    - 140-110: 8 points
    - 110 seconds or less: 10 points
  - The highest-scoring team: 5 additional points will be awarded to the final grade.
  - This team's spaceship design, control system code, and video will be made public.

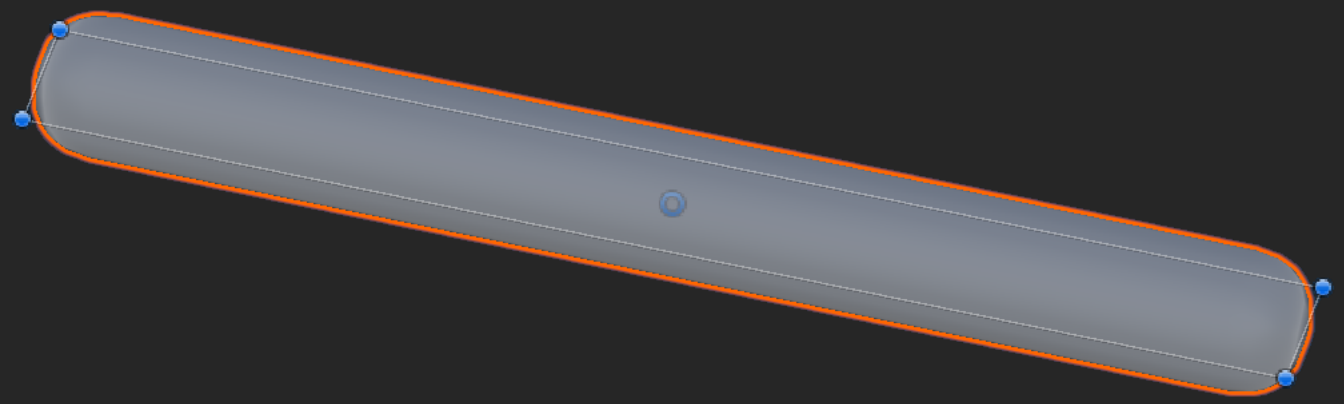
Evaluation criteria

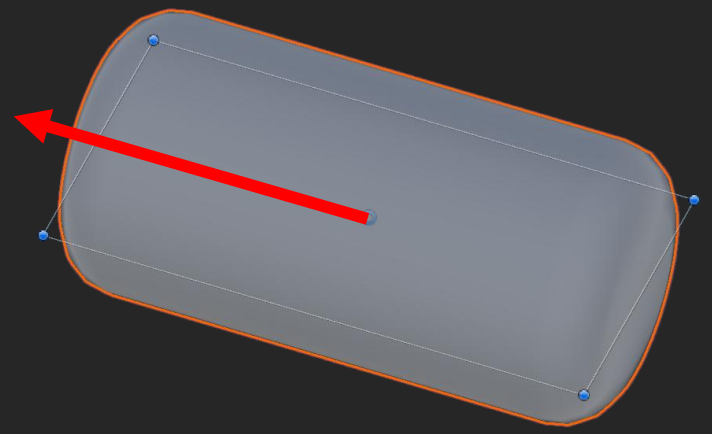
- Presentation (15 minutes, 15 points, Individual score)
  - Performance record, rocket introduction
  - Each team member's presentation of their role **(All team members must present)**
    - Controller design
    - Rocket design
    - Control system design
  - 10/21 (Tue): 6 teams
  - 10/23 (Thu): 5 teams
  - **If you are not prepared for your presentation, you must attend the additional session on 10/23 from 19:00 to 22:00 for an additional presentation.**

Spacecraft

Spacecraft that violate the constraints will have their performance report disqualified (0 points).

- Body, Engine, brake, servo motor



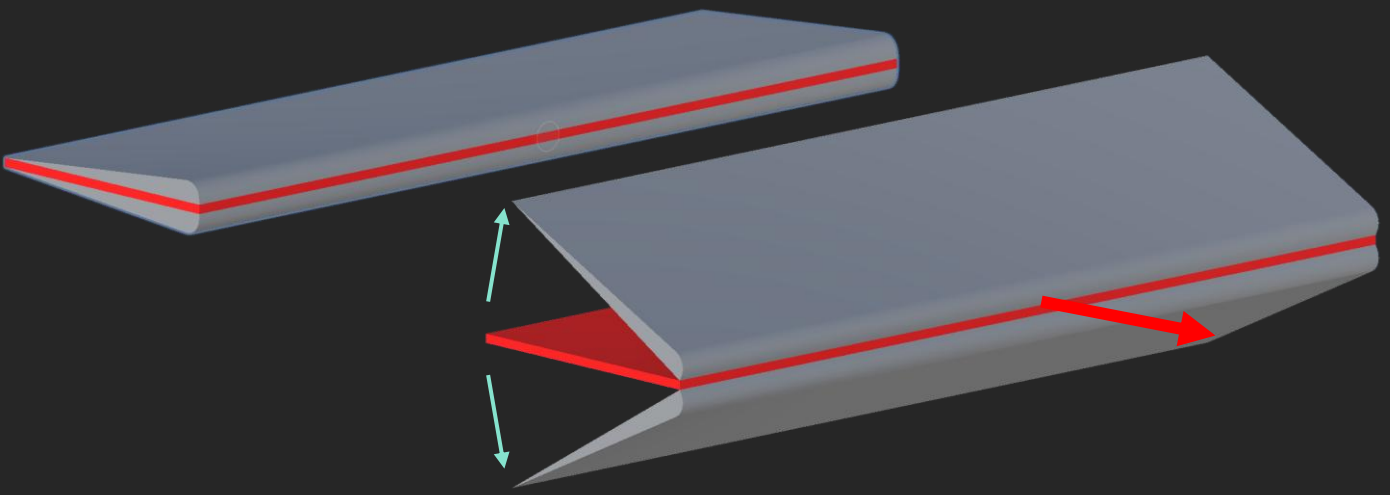


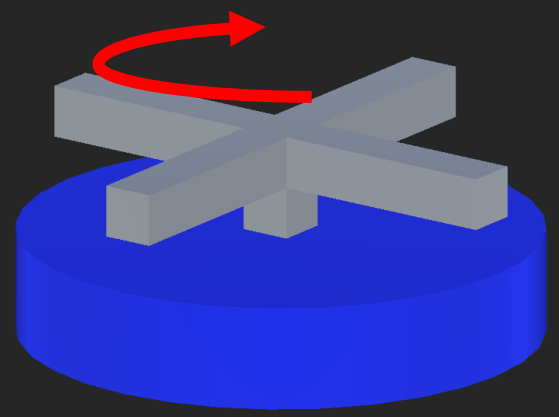
Body
Every part must be child object of body.
Can use only 1 body
Dimension: 2 * 0.25 * 0.25 m bar
Mass: 10kg
Total spacecraft size must fit in 9 * 9 * 5m box.
(Same as a size of stating zone)

Engine (float 0.1~1)
Add thrust to spacecraft
Dimension: 1 * 0.5 * 0.5m cylinder
Mass: 1kg
Variable thrust: 5~60N
Stronger thrust > Bigger Size (Automatically set)
Total engine thrust must be under 60N

Spacecraft

- Body, Engine, brake, servo motor



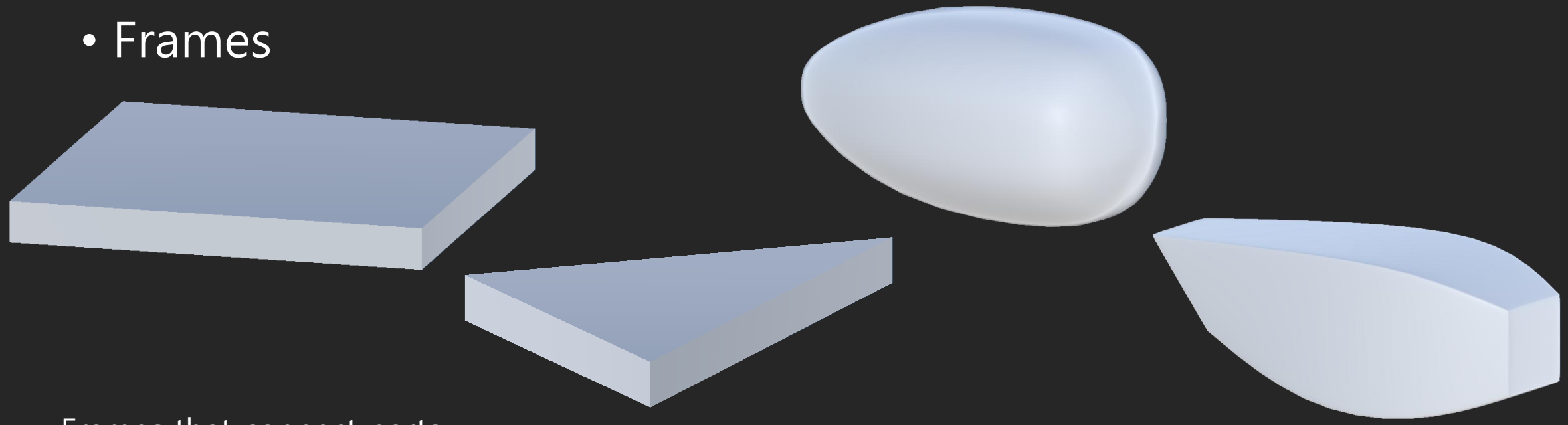


Brake (float 0~1)
Provides a resistance force of -10 * velocity^2 (N)
Cannot decelerate below 0.1 m/s
Dimension: 1 * 0.3 * 0.06 m plate
Mass: 0.1kg
Do not change the size or coefficient of the brake

Servomotor (float 0~360)
Rotate a child object to a desired angle.
Dimension: 0.1 * 0.1 * 0.04 m cylinder
Mass: 0 kg
Constant angular velocity: 600 deg/s
Do not change the angular velocity of the motor.

Spacecraft

- Frames

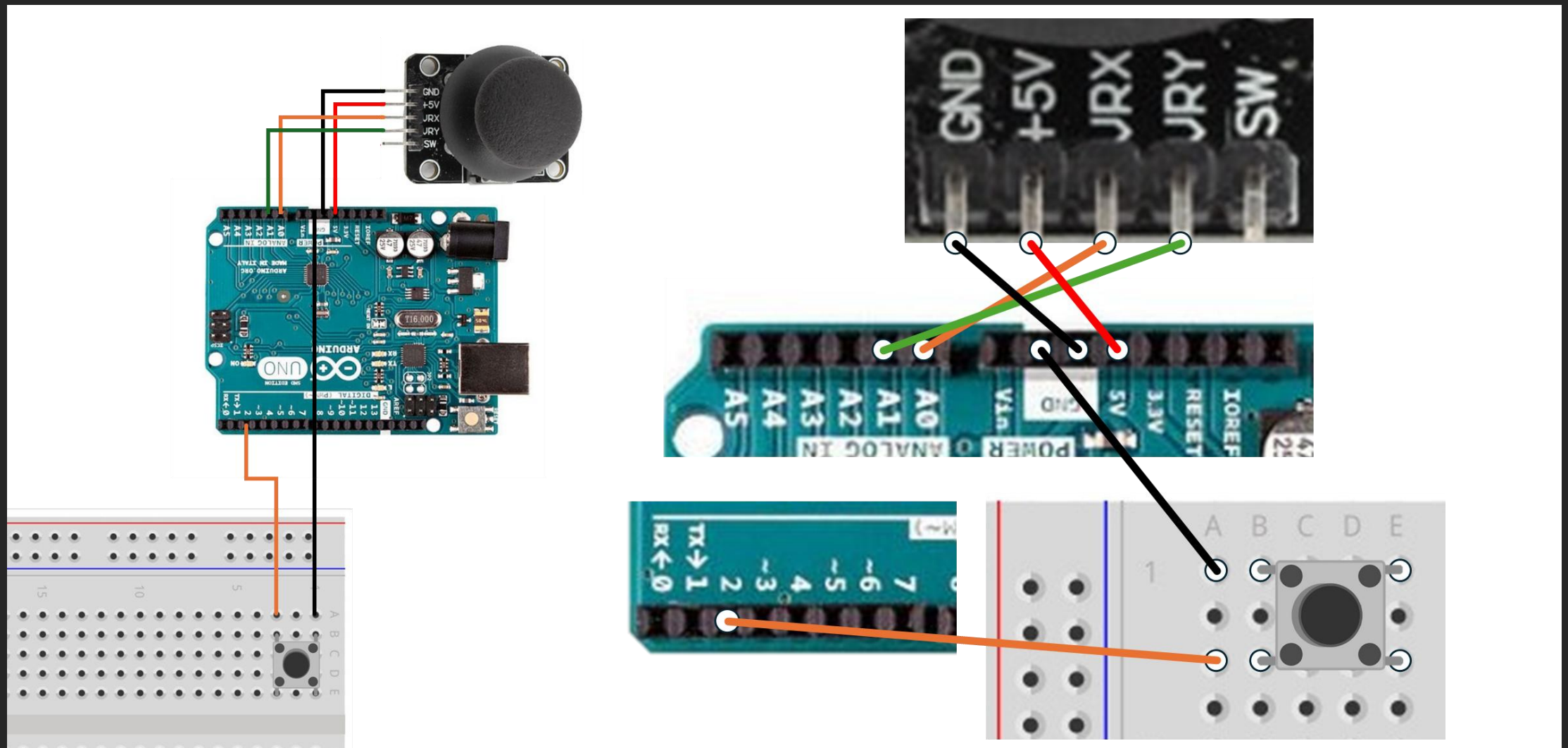


Frames that connect parts.
Free to scale and use for design purposes.
A completed spacecraft must not have any disconnected parts.
Except for the frame, no parts can overlap.
Design so that interference does not occur even when the part is rotated by a servo motor.
Dimension: Free
Mass: 0 kg

Controller

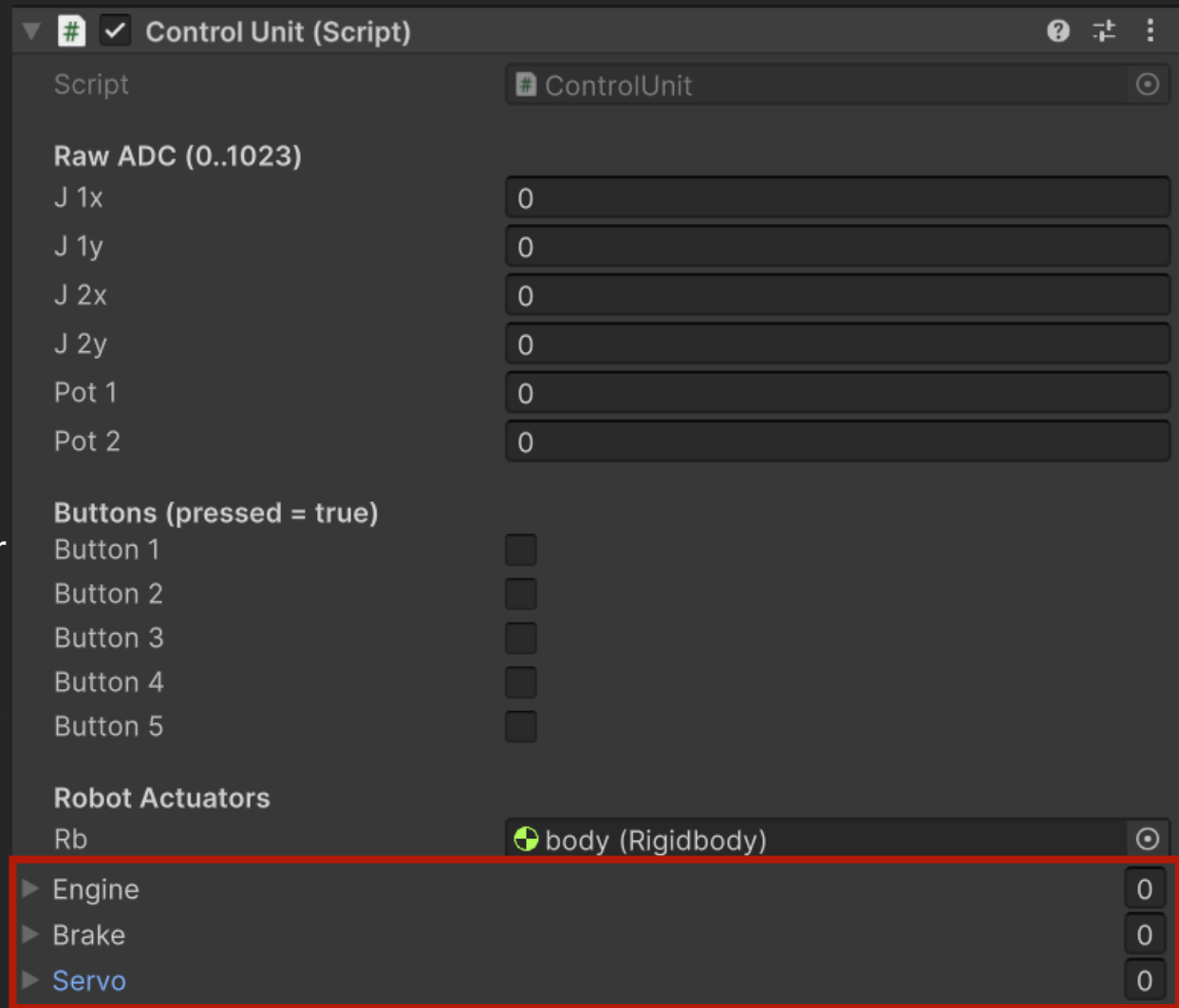
- Build your own controller using Arduino.
- The following input components are provided for project 1:
  - 5 switches
  - 2 joysticks
  - 1 rotational potentiometer
  - 1 breadboard (and usb cable)
  - Jumper wires
- You can build them using items around you (please consider sustainability).
  - Recycled materials, cardboard, plastic bottles, etc.

Example Controller



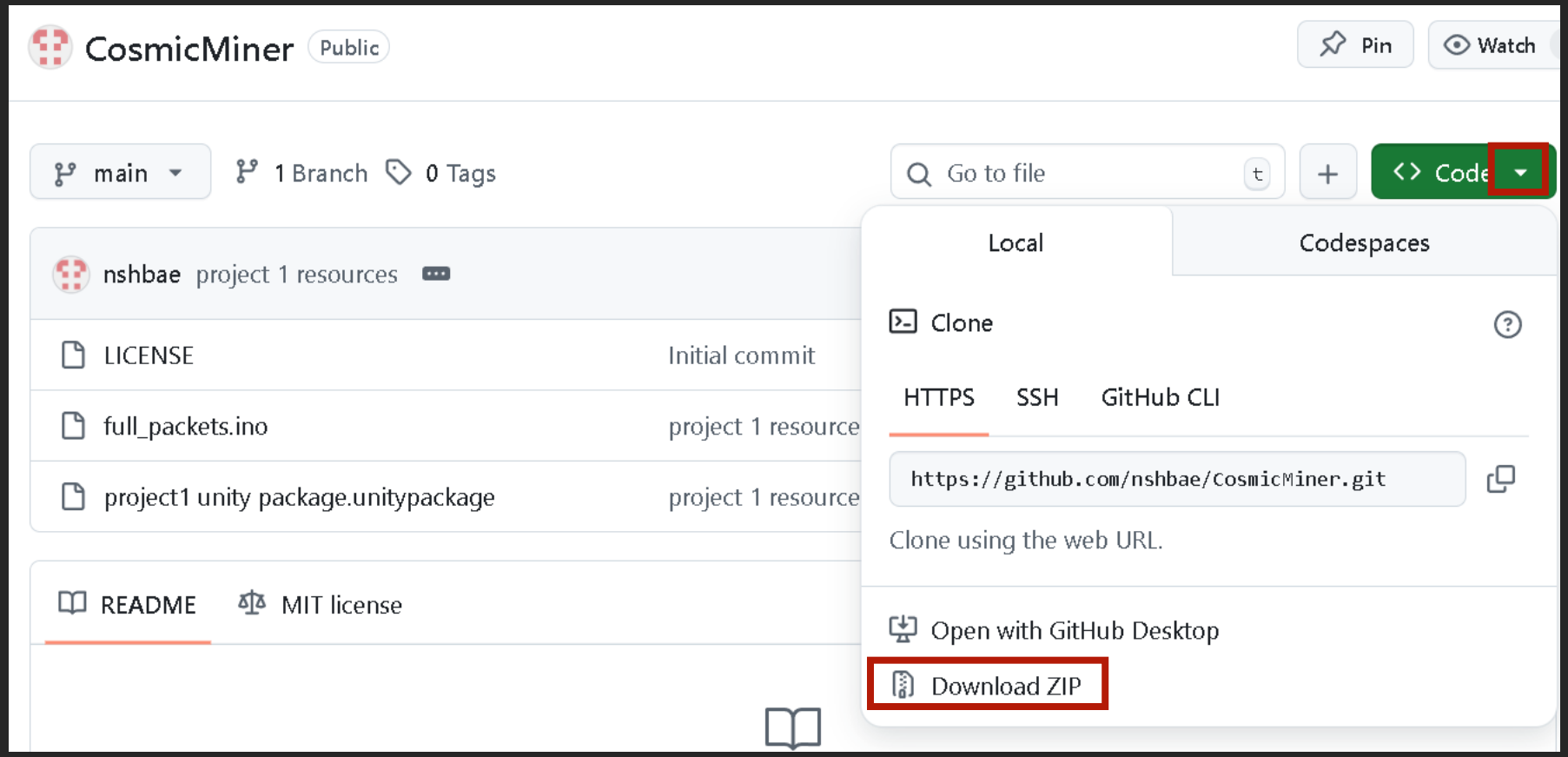
Control System

1. Add the engine, brake, and servo objects of the designed spacecraft to the list by dragging and dropping them.
2. Edit the ControlUnit.cs file to map the controller's input and control values.
   - Only scaling, offset, and logical operations are allowed for mapping.
   - Controls that use the spacecraft's position and rotation as input are prohibited in Project 1.
   - If you have any questions about whether your desired mapping is permitted, please ask to TA.



Let’s start learning Unity Engine

- https://github.com/nshbae/CosmicMiner



Unity Basics

Scene – A virtual space where simulations are run.

Game object
Objects placed in the scene

Component
Properties that make a Game Object work

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Properties that make a Game Object work

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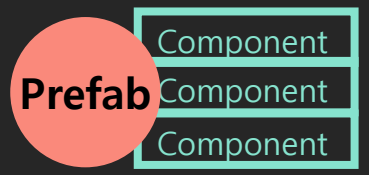
Properties that make a Game Object work

Assets – All resources (files) used in the game.
Scripts, images, models, sound effects, background music, scenes, data, saves, etc.
They are saved as files in a folder called Assets within the project folder.



Prefabs - Game object assets with pre-configured components for easy placement in a scene.
Pre-saved prefabs can be easily used by dragging and dropping.

Window, Layout

Hierarchy
Displays the parent relationships between objects within the scene.

Project
Folder, Asset, and Resource Management

Inspector
Components of the selected object. Options of the selected asset, etc. View/edit properties

Scene
A screen where you can place objects and freely look around.

Game
The screen the player sees

Log
Errors, warning messages, log messages, etc. are displayed.

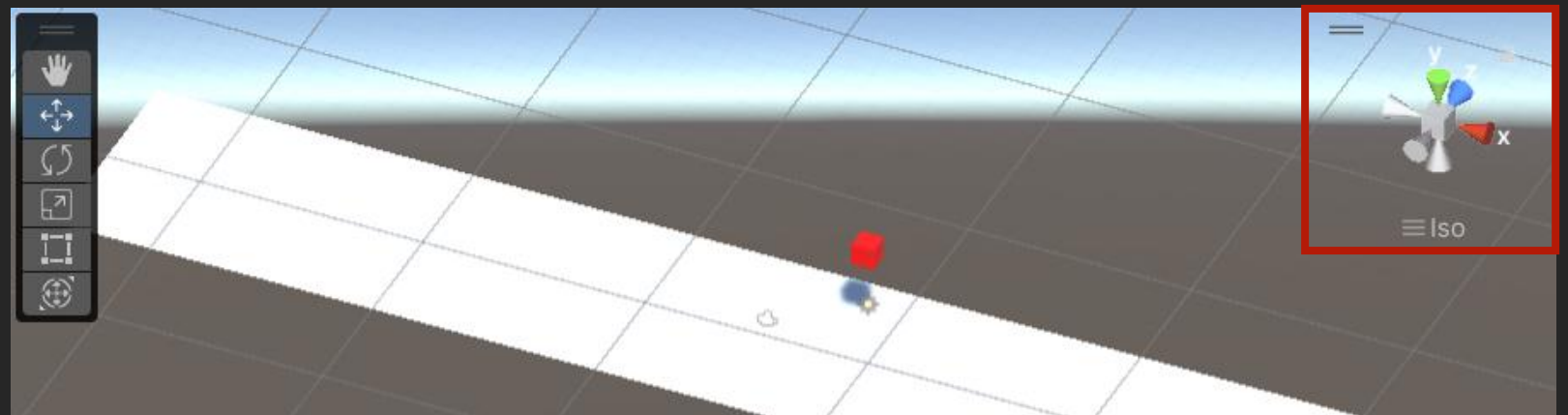
The arrangement of windows is called a layout. Currently, it is a 2 by 3 layout. If you want to change the layout, go to the menu bar - Window - Layouts and select the desired layout.

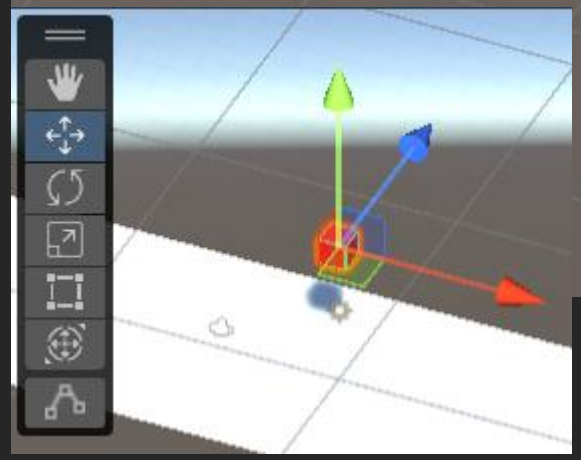
Scene

Move the viewpoint (shortcut Q)
Right-click to rotate the viewpoint
Click the mouse wheel to pan

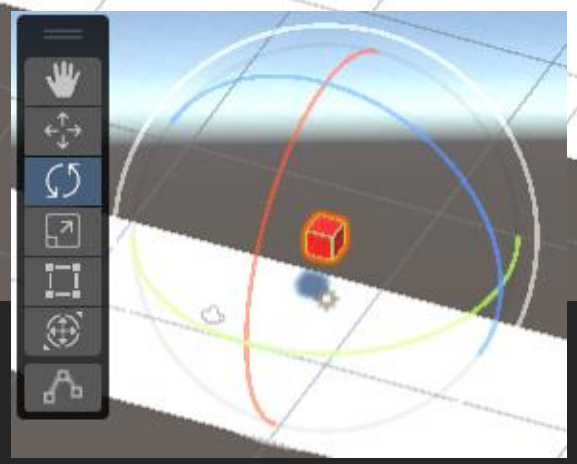
Click on each axis to change the viewpoint.
Click on the cube to change the perspective.
- Perspective (Persp): First-person, with perspective.
- Orthographic (Iso): Parallel view, without perspective.

Double-click on an object in the Hierarchy to check its position.

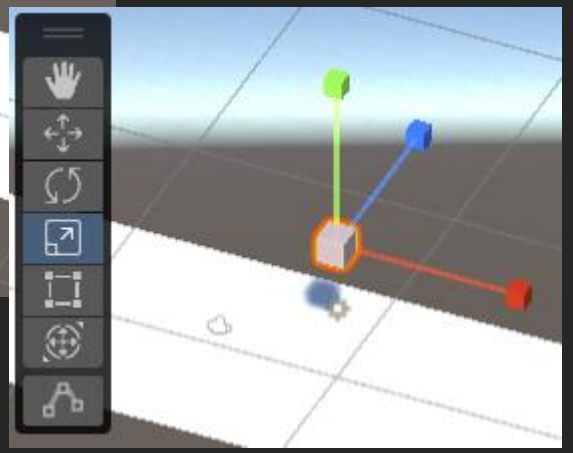




Move position (shortcut W)
Grabbing the tip of an arrow will move only along that axis.
Grabbing the square between two arrows will move only on that plane.

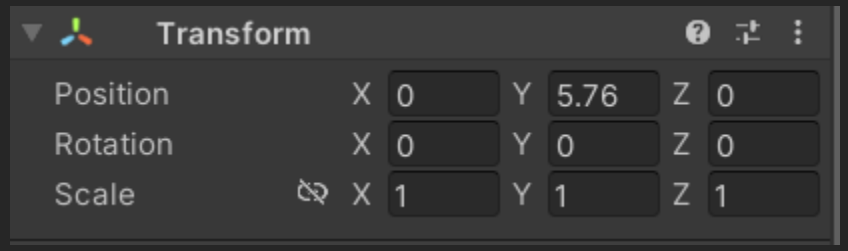


Rotation (shortcut E)
Click on a colored circle to rotate it only along that axis.
Click on any point on the sphere to rotate it along all axes.



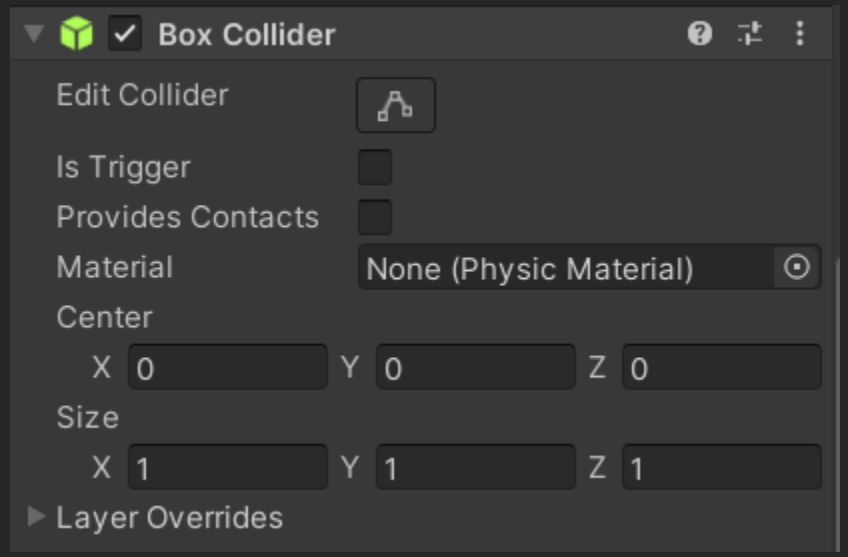
Scale (shortcut R)
Clicking the cube at the end of an axis will scale only along that axis.
Clicking the white cube in the center will scale equally along all axes.

Basic Components



Transform

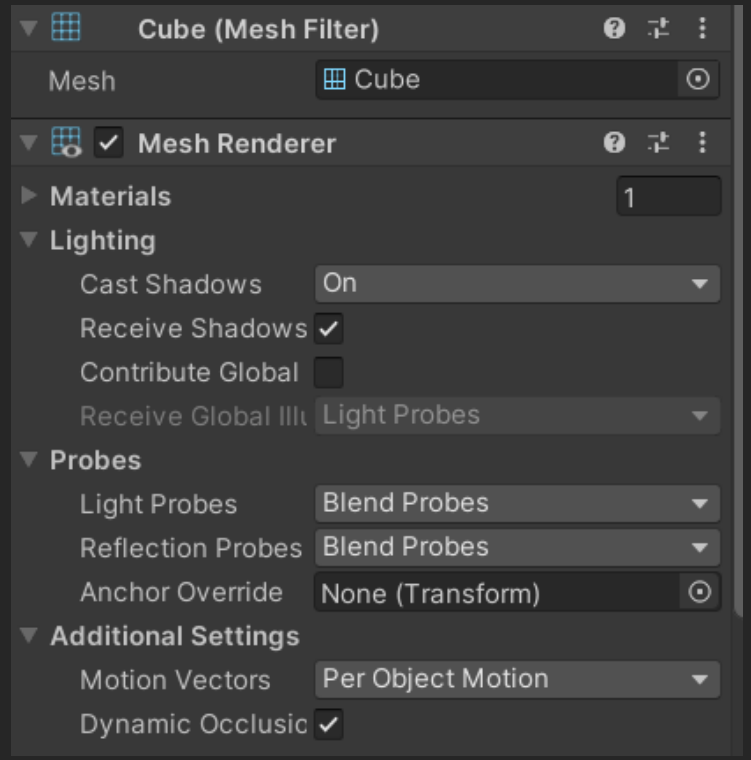
Components are the basic properties of all objects. They include position, rotation, and scale. You can edit these values by directly entering them, or by pressing X, Y, and Z and dragging the mouse.



Collider

A component that specifies the collision area. It is indicated by a green border, and can also specify an invisible collision area (this determines the start and end points of the track).

Basic Components



Rigidbody

Components that are the subject of physics calculations

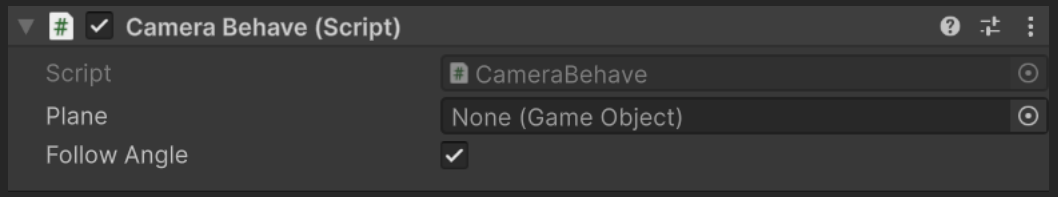
You can configure mass, drag, angular drag, center of gravity, and whether gravity is applied.

You can select constraints to fix position or rotation.

Mesh Filter, Mesh Renderer

You can configure which mesh (3D model) to display and how lighting and shadows will be rendered when displaying it.

Basic Components



Script Components

A component controlled by a script. Its name is displayed as the name of the script.

Variables declared as public within a script can have initial values set within the editor.

(Objects or components can be added by dragging and dropping them from the hierarchy.)